# Advanced Geospatial Search Strategy for Ryan Shtuka

## Leveraging LiDAR and Anomaly Detection at Sun Peaks Resort




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# Executive Summary: The Search Pivot

## The Incident

20-year-old Ryan Shtuka vanished February 17, 2018, at 2:10 AM in Sun Peaks, BC.

## The Barrier

22 centimeters of fresh snow and -26°C temperatures acted as a natural eraser, burying potential tracks, neutralizing K9 scent trails, and blocking infrared drone cameras.

## The Pivot

Transitioning the 7-year investigation from traditional boots-on-the-ground visual sweeps to advanced bare-earth topographical analysis.

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# The Geographic Challenge




## Canopy Obscuration

Dense tree cover blocks standard aerial photography and line-of-sight searches.

## Tree Wells & Gullies

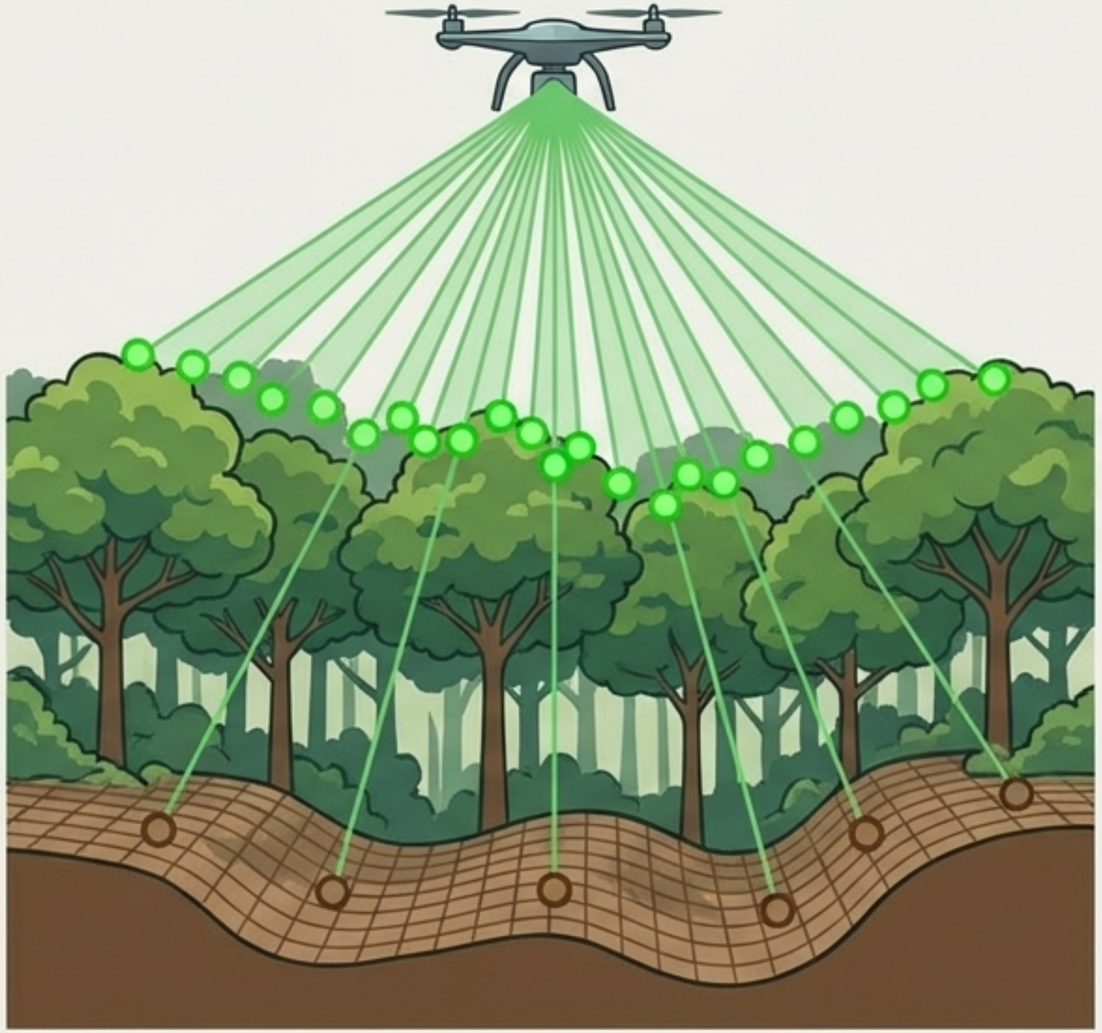
Natural hazard zones where a person could succumb to the elements completely unseen.

## Surface Evolution

7 years of new vegetation, leaf litter, and deadfall have fundamentally altered the terrain, rendering visual sweeps ineffective.

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# LiDAR as the Catalyst



Digital Surface Model

Bare Earth

## The Science

Light Detection and Ranging (LiDAR) uses pulsed lasers to strip away foliage and map exact ground elevations.

## The Advantage

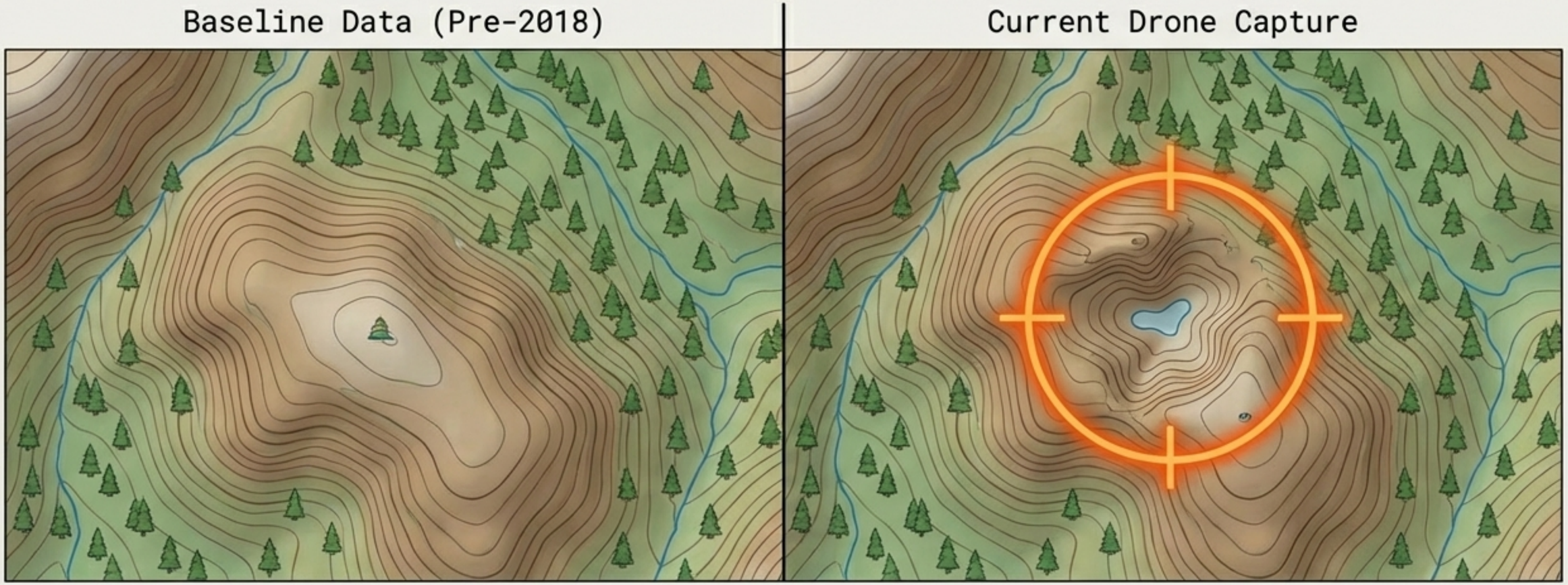
Penetrates 7 years of dense overgrowth to reveal the true topographic surface hidden beneath.

## Data Availability

Leveraging the National Elevation Data Strategy and LidarBC's High-Resolution Digital Elevation Models (HRDEM) covering the region.

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# Historical LiDAR Comparison




## Baseline Establishment

Utilizing historical Canadian elevation data to understand the natural contours of Sun Peaks prior to 2018.

## Delta Identification

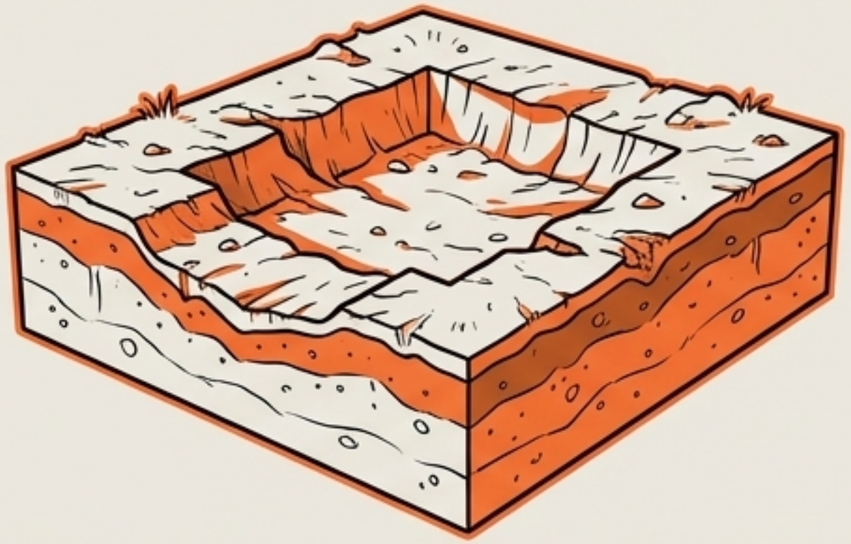
Overlaying new, high-density drone LiDAR scans to isolate post-2018 topographical changes.

## Objective

Eliminate static geographical features to focus exclusively on new terrain disturbances.

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# Differentiating Anomalies in the Point Cloud

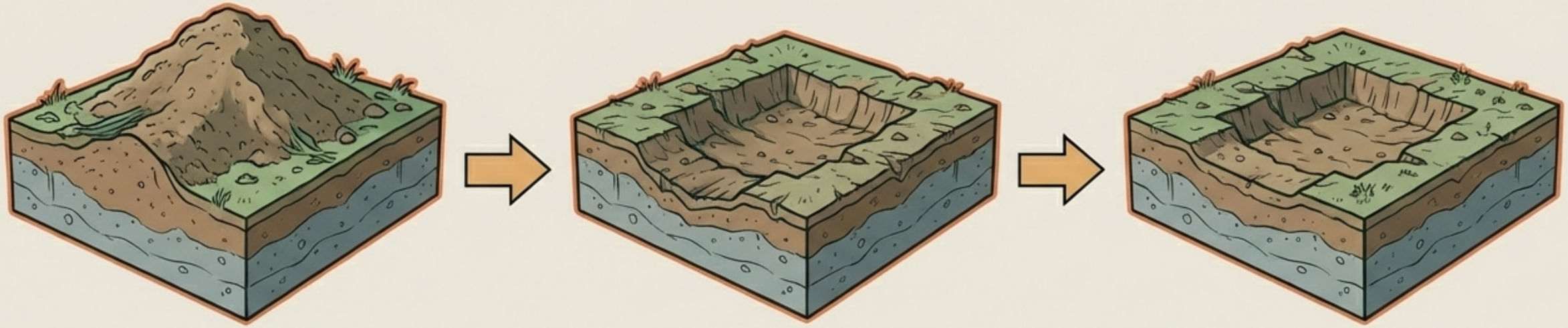
| Organic Anomalies | Manmade & Disturbance Anomalies |
| --- | --- |
|  |  |
| **Natural Processes:** New vegetation growth, dry leaf litter, natural deadfall accumulation. | **Clandestine Signatures:** Irregular soil settling, unnatural depressions, localized disturbance mimicking natural processes. |

**Critical Task:** Utilize geospatial software to filter organic noise and reveal manmade geometric irregularities.

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# The 3-Phase Disturbance Signature

Based on forensic terrestrial LiDAR studies, isolated elevation loss surrounded by undisturbed ground is a primary target indicator.



### Phase 1: Elevation Gain

Initial disturbance and addition of mass.

### Phase 2: Elevation Loss

Soil settling, compaction, and decomposition.

### Phase 3: Stasis

Uniform elevation change across the disrupted surface.

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# Macro Map Analysis: The 5km Radius

**Macro View**
Sun Peaks Village & Environs

**The 5km Constraint**
Statistical limit for standard wilderness disappearances, extending to the top of the ski hills.

**Focus Shift**
Moving from the expansive 5km net to highly localized, data-driven micro-zones.




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# Target Zone A: Burfield Drive & The Ski Out

**Zone A Details**
The immediate perimeter of the 2:10 AM departure from the house party.

**Priority Indicators**

- Deep snowbanks
- Steep roadside drop-offs
- Dense tree clusters immediately adjacent to the walking route




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# Target Zone B: Fairways Drive Periphery




**Hypothesis**
Investigating the unconfirmed 1:30 AM sighting. Potential route taken to grab food in the village before the 2:00 AM business closures.

**Priority**
Unchecked roadside ditches and tree lines where industrial snowplows operated that morning.

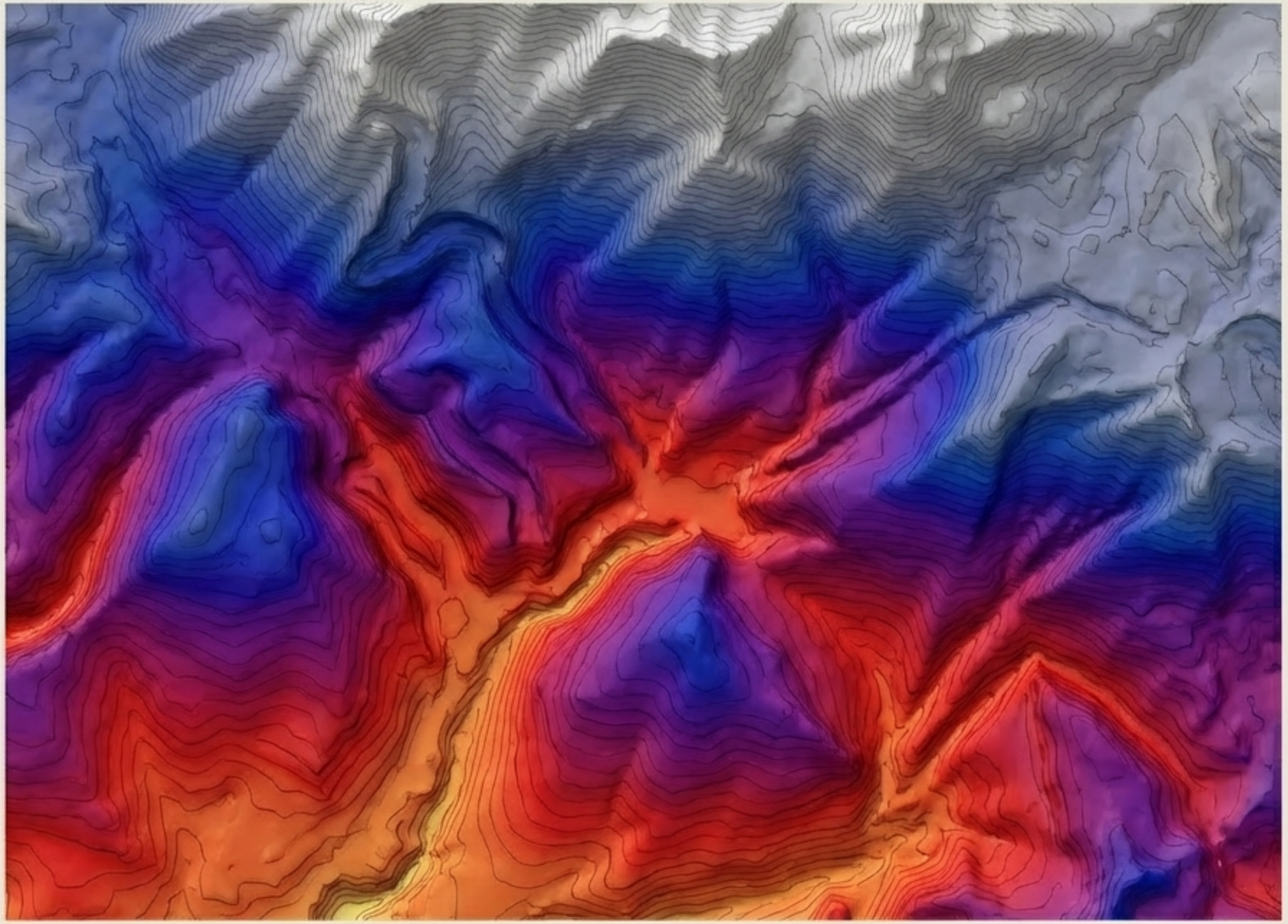
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# Target Zone C: High-Risk Terrain

**Environmental Traps**
Focusing on areas where an intoxicated or disoriented individual in -26°C weather would naturally seek shelter or accidentally fall.

**Search Priorities**

- Unsearched steep drops
- Hidden creek beds
- Dense groves known for hazardous tree wells



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# High-Priority Drone Waypoints



**Waypoint Alpha (Burfield Ridge)**

50° 53.58' N, 118° 38.67' W

**Waypoint Beta (Fairways Approach)**

50° 53.82' N, 118° 37.41' W

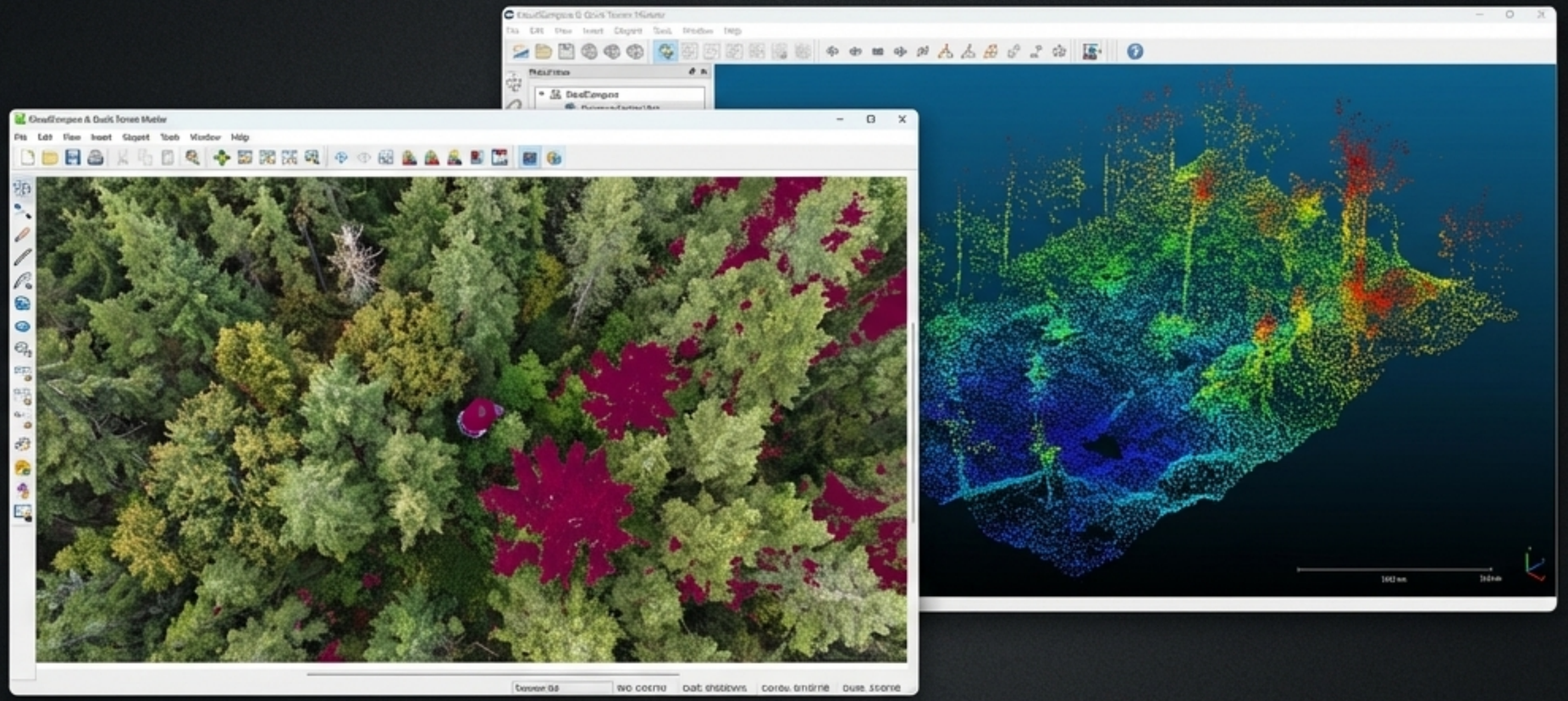
**Waypoint Gamma (Ski Out Ravine)**

50° 53.15' N, 118° 39.02' W

**Action:** Deploy drone-based LiDAR to these precise vectors for high-density point cloud generation.

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# Software & Data Integration



## CloudCompare & Quick Terrain Modeler

For 3D point cloud visualization, baseline matching, and identifying Phase 2/Phase 3 geometric depressions.

## Color-Matching Algorithms

Proprietary software to scour localized drone photography for specific RGB values matching Ryan's burgundy ball cap, blue coat, and dark jeans.

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# Required Resources & Logistics



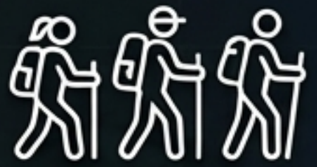
## Hardware

Drone-based LiDAR systems (e.g., Emesent Hovermap) capable of penetrating thick canopy.



## Software & Data

High-end processing units for Cloud Optimized Point Cloud (COPC) files and access to LidarBC historical datasets.



## Ground Truth Teams

Advanced volunteer teams equipped with avalanche probes and precise GPS receivers to investigate identified anomalies.



## Coordination

Direct data handoffs and reporting protocols with the Kamloops RCMP.

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# The Mission Continues

**We will never stop searching.**

Seven years of growth have changed the mountain, but LiDAR removes the veil. By shifting from visual sweeps to geospatial anomaly detection, we target the unseen.

## Immediate Next Step:

Acquire current LidarBC sets, initiate drone mapping of Zones A & B, and deploy ground-truthing teams to verified coordinates.

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